

CUPERTINO POLICE DEPT. INVESTIGATIONS UNIT

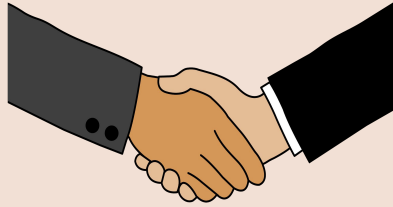
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TYLER STOEN, ZARINA YUNIS

A d



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THE PEOPLE OF CUPERTINO V. KELLY AND ALEX



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THE PEOPLE OF CUPERTINO V. KELLY AND ALEX



256 PHOTOS BY ALEX



20 UNLABELED PHOTOS



229 PHOTOS BY KELLY



24 UNLABELED PHOTOS

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GOALS

01

TRAIN MODEL TO CLASSIFY KELLY VS ALEX

02

TRAIN EXPLANATORY MODEL

03

COMBINE THESE 2 MODELS



MAIN GOAL:



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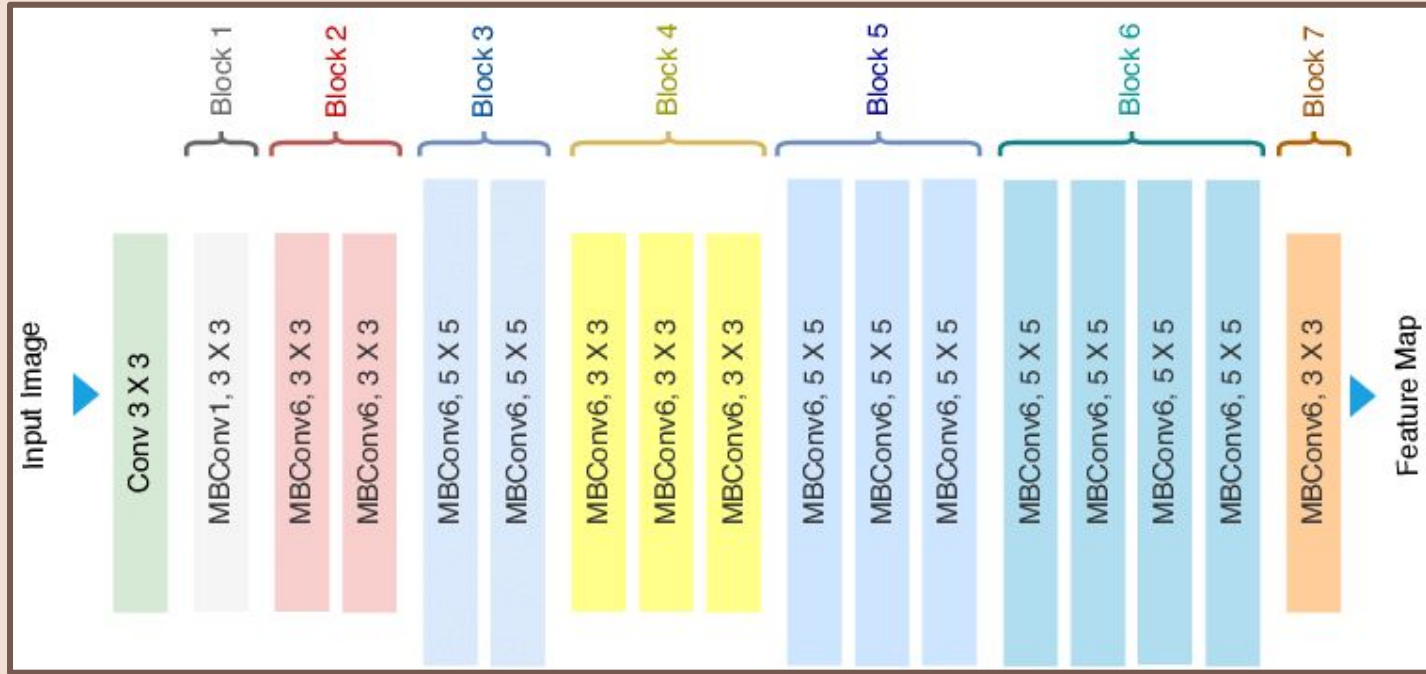


01 PHASE 1

Image Classification
with Neural Nets

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MODEL: EFFICIENT NET - B0 ARCHITECTURE

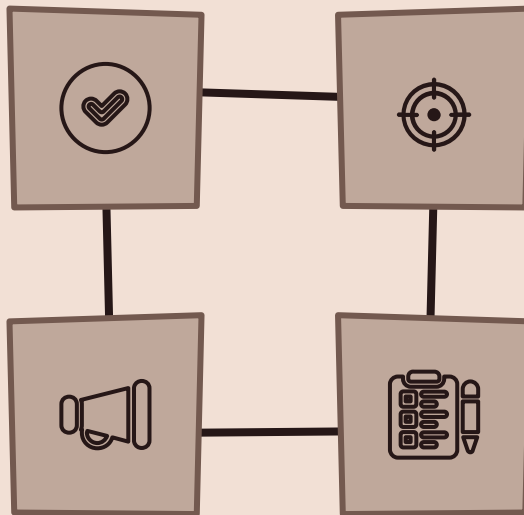


Source: [OpenGenus](#)

RESULTS



ACCURACY
0.9178

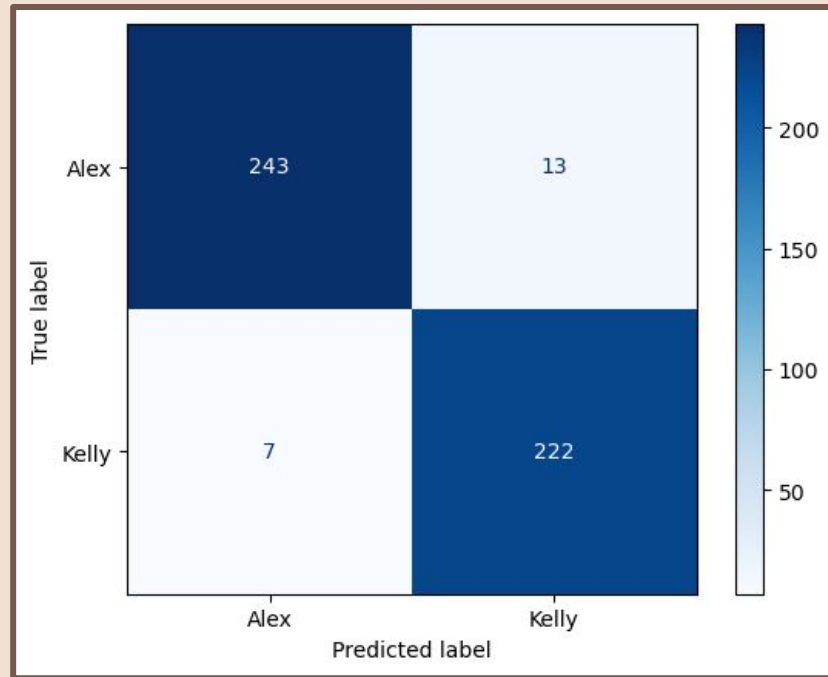


PRECISION
0.8889

RECALL
0.9412

F1-SCORE
0.9143

CONFUSION MATRIX





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02

PHASE 2

Extracting important
features

5

FEATURES

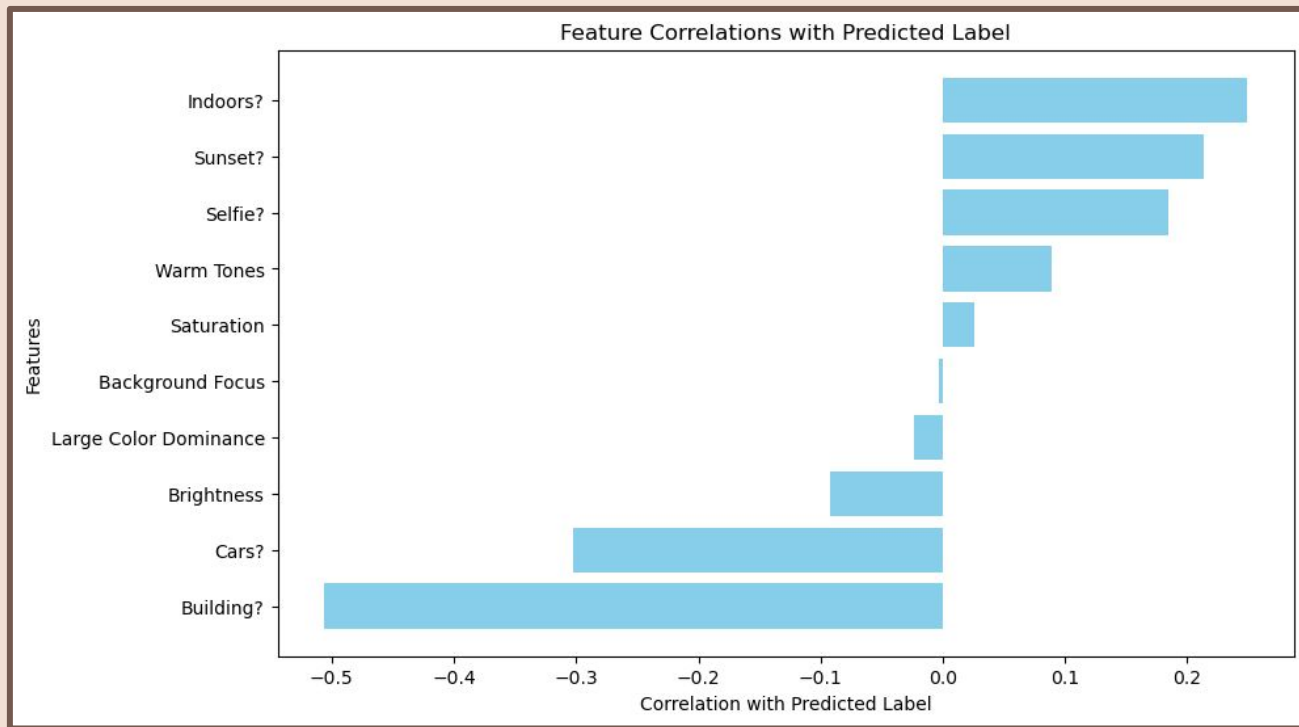
COMPOSITION

- High Saturation
- High Brightness
- Single Color Dominance
- In Focus
- Warm Tones

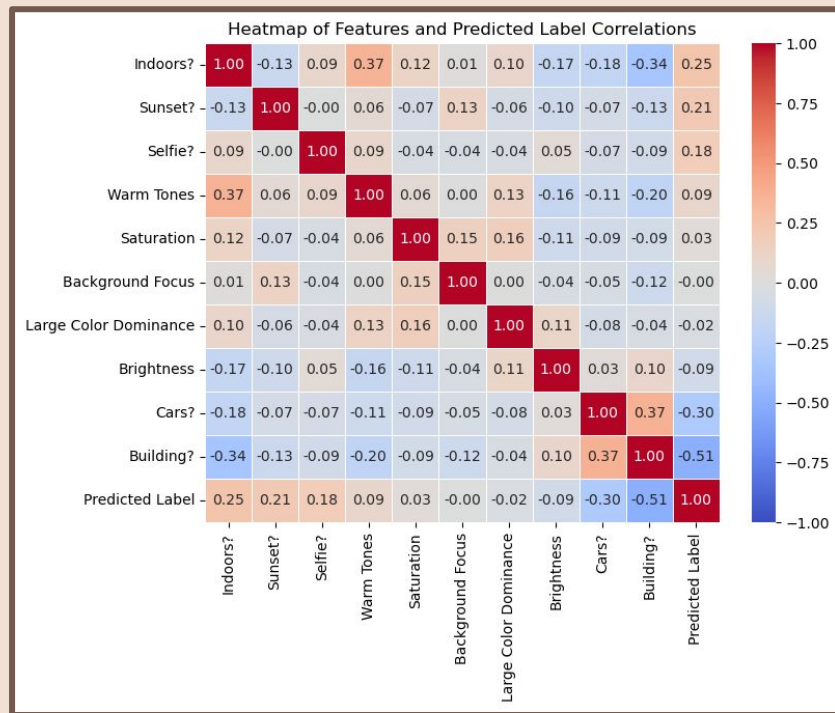
CONTENT

- Indoors
- Selfie
- Sunset
- Buildings
- Cars

CORRELATIONS



CORRELATIONS

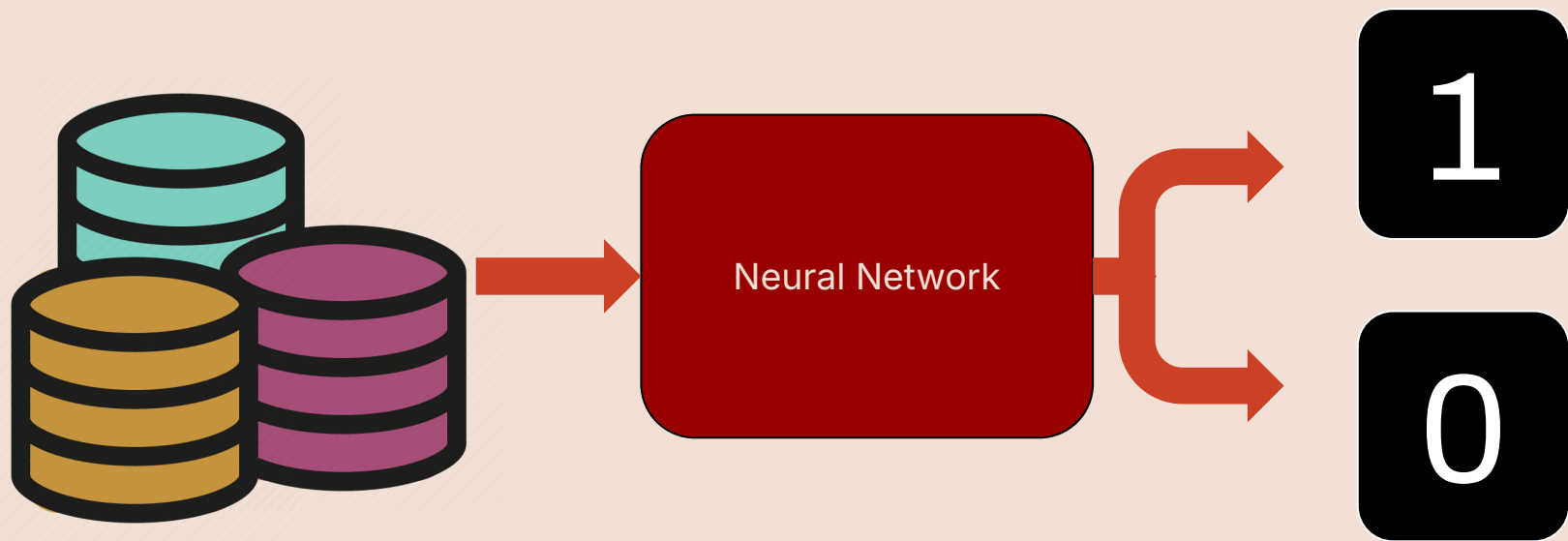


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EXPLANATORY MODEL



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68.5%

Validation Accuracy

72.6%

Test Accuracy



A

03

PHASE 3

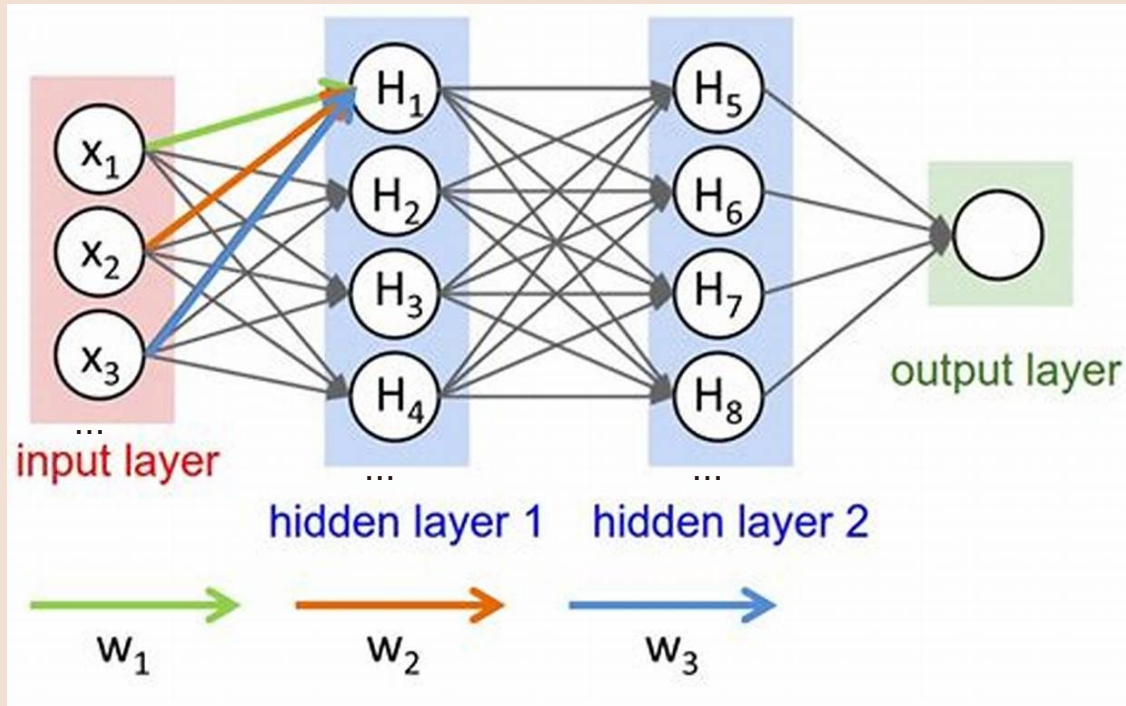
Combine Models from
Phase 1 and Phase 2

Q



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ENSEMBLE MODEL



91.8%

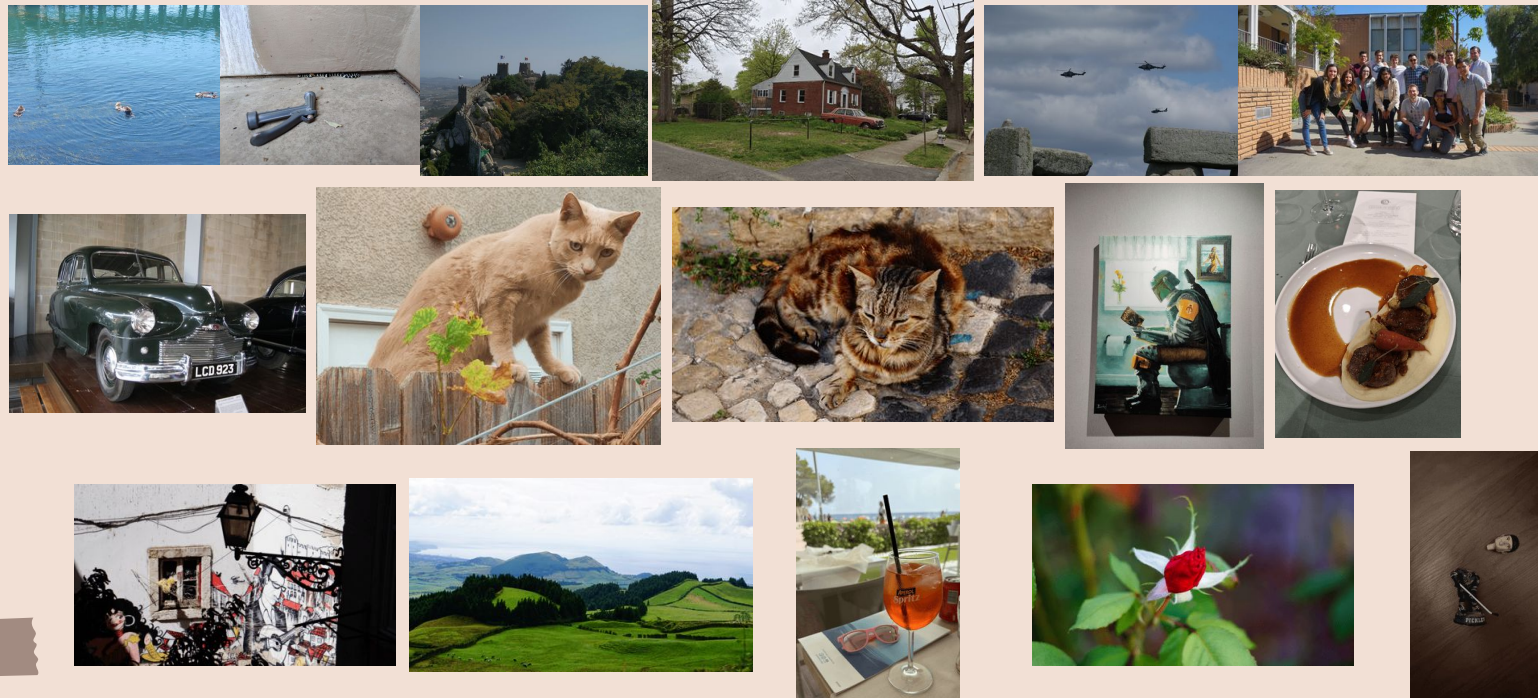
Validation Accuracy

95.9%

Test Accuracy



ALEX



L

KELLY





**DO THE RIGHT THING
PUT THESE TWO BEHIND
BARS**



**THANK
YOU**
